

EE1102 : Introduction to Programming
Feb-March 2022

Things to Know for this Assignment

All writing and reading requires including `stdio.h`:

```
#include<stdio.h> /* this is a comment line (can be on separate line too) */
```

Basic structure of a C program:

```
#include....  
int main(){ /* start of the main function. All execution starts here */  
    .... /* code */  
    return (...); /* return value to the operating system */  
} /* end of the main program */
```

To write something, use `printf("...")`.

```
printf("This is a line\n");  
This is a line
```

To print multi-line output

```
printf("This is\na line\n"); /* each \n starts a new line */  
This is  
a line
```

To declare a variable and assign it a value

```
int a;  
a=4;
```

To write something that includes an integer variable (place holder for integer is `%d`)

```
printf("The value of a = %d\n",a); /* output printed, with %d replaced by value */  
The value of a = 4
```

Addition of variables and constants

```
Int b=10;  
Int c=15; /* can give the variable a value inside the declaration itself */  
Int d,e;  
d=a+b+c; /* addition of variables */  
e=a+4+c; /* addition of variables and constants (here 4 is a constant) */
```

Sample Assignment

Problem 1. Write a C program called Hello.c that prints the message below.

```
Hello.  
**This is my first program for the EE1102 lab.**
```

Problem 2. Write a C program called sum.c that prints the sum of 5 integers in the format shown in the example below. Use variables of type int. Try changing the values for each variable and check if the output is correct.

Example: If the 5 integers you want to use are 10, 25, 5, 5, and 3, it
should print $10 + 25 + 5 + 5 + 3 = 48$

Problem 3. Write a C program called print_my_details.c that prints your roll number in the first line and your name in the second line.

Example: If a student's roll number is EE20B888 and name is Abc Xyz, the output printed when we run the program should be

```
EE20B888  
Abc Xyz
```

Problem 4. Write a C program called `sum_of_squares.c` that prints the sum of squares of 3 integers in the format shown in the example below. Use variables of type `int` .. Try changing the values for each variable and check if the output is correct.

Example: If the 3 integers you want to use are 10, 25, and -5, it should print

A = 10; B = 25; C = -5;

$A^2 + B^2 + C^2 = (10)^2 + (25)^2 + (-5)^2 = 750$